CSH interview: final preparation pack

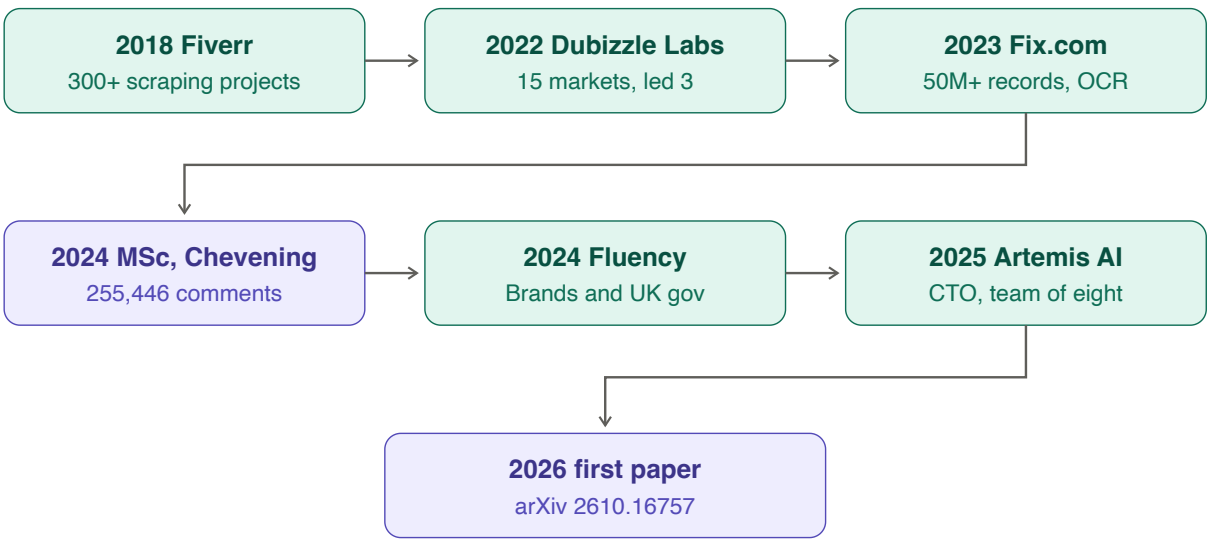
Preliminary interview **Tue 20 Oct 2026, 4:40–4:55 pm UK = 8:40 pm Lahore** · Zoom 843 5332 6447 · Graduate Program in Complexity Science, Digital Innovation School · interviewer contact: Carrie Cowan, Head of DIS

1. How this interview works (researched)

- **Format:** about 10 minutes of standard questions on your prior experience and future interests, plus a few minutes of instructions; in English on Zoom. CSH does not share questions in advance and says there is no need to prepare or study.
- **It counts:** the recording and transcript become part of your application package and are read by the Selection Committee alongside your written answers. So keep every fact consistent with what you submitted.
- **What the committee looks for** (CSH FAQ): fit between your background and CSH research, and "potential for independent and rigorous scholarship in the area of complexity science".
- **After this:** Stage 1 committee review → Stage 2 panel interview on Zoom, 7–14 January 2027, where you present a paper chosen by the committee and answer questions on it → Stage 3 finalists present their research ideas at CSH, 15–16 February → offers by mid-March. Referee letters must arrive by **1 November**.
- **What good answers look like** (PhD interview guides): specific, concrete examples of what you did; a clear reason for a PhD that comes from a real experience, not a slogan; naming their work that moved you; and showing gaps as things you are already learning.

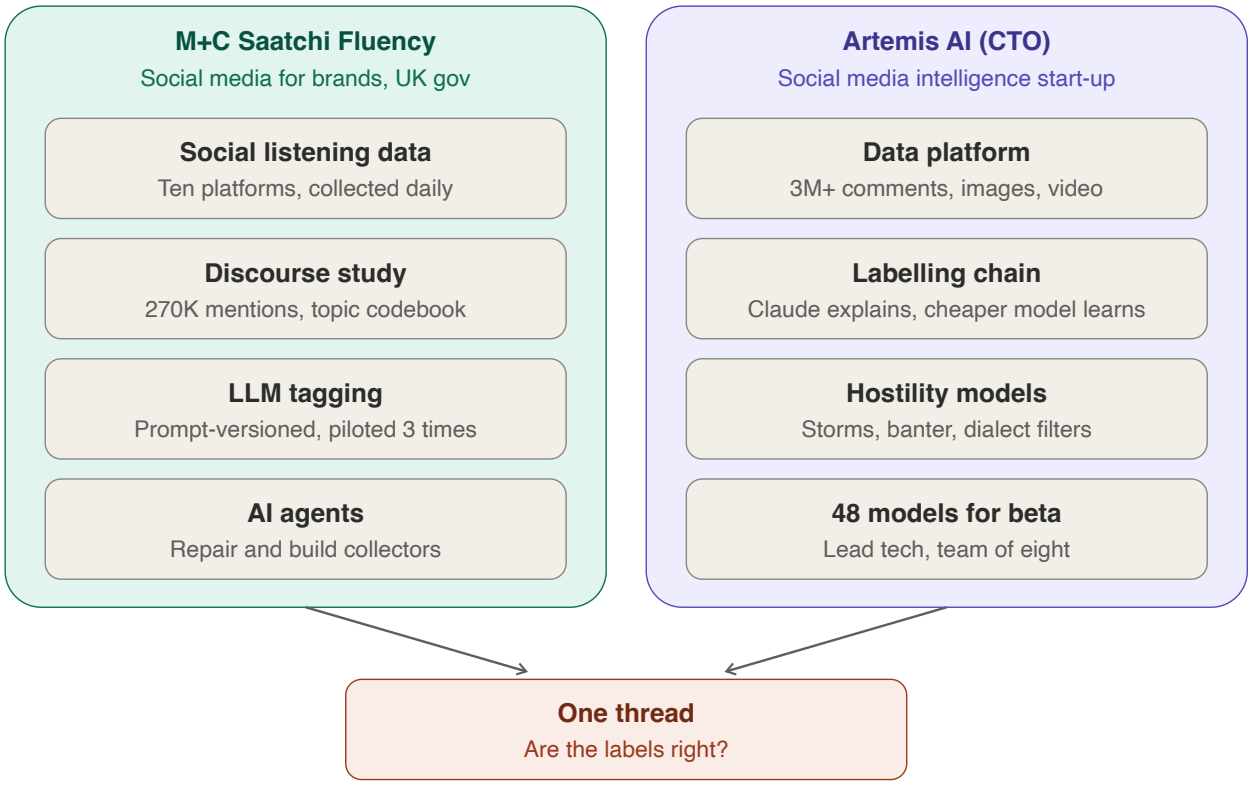
Three things to land in any answer.
Question: do machine and LLM labels of online text mean the same thing across platforms, models and time, and how do we correct comparisons when they do not?
Proof: your first paper, with Dr Ella Haig, on arXiv since 14 October (2610.16757), which grew from your dissertation's circular validation.
Edge: eight years collecting and labelling social media at scale (Dubizzle, Fluency, Artemis AI); data is never your bottleneck.

2. Your path, 2018 to 2026



Teal = industry, purple = academic. Fiverr taught scraping and clients; Dubizzle multilingual data and leading people; Fix.com data quality at scale; MSc research method and honest validation; Fluency social media analysis that decisions rely on; Artemis AI building and evaluating models; the paper turned a weakness into research.

3. What you did at Fluency and Artemis AI



M+C Saatchi Fluency (London, since December 2024). Fluency is the social and web intelligence side of M+C Saatchi, and its reports go to global brands such as Amazon, Ford, Nike and Reckitt, and to UK government communications teams. Your work there has two halves.

- **Research and analysis.** You were an analyst on a twelve-month study of public discourse: 100,845 Brandwatch mentions and 170,542 Reddit mentions. Instead of hand-reading them, the team built a codebook from BERTopic topics, then used an LLM to tag every mention. The study used two research habits you now treat as standard: every tag carries a hash of the exact prompt version that produced it, so results can be traced and repeated; and the prompt was run three times on the same 500 rows before the full run, to see whether the labels were stable. In your portfolio you called this treating a prompt as an instrument whose version must be recorded. It is the closest you have come to studying narratives at scale.
- **Data and AI engineering.** You collect from ten platforms every day, feeding a governed master table. When a forum changes its layout, a crawler usually breaks silently: the series does not stop, it just thins, which quietly biases any trend. You designed a self-healing AI agent that diagnoses the break, rewrites the collector, and tests it on the live site before anything ships, plus a builder agent that creates new collectors. Both follow fixed rules, and a person approves every change.

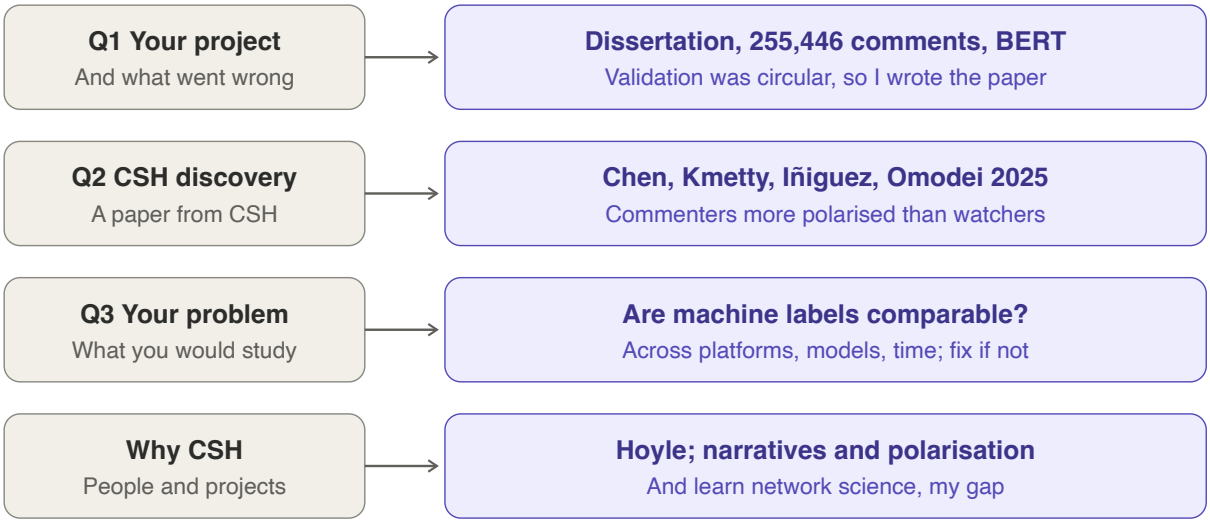
Artemis AI (co-founder and CTO, since 2025). A social media intelligence start-up helping creators and businesses understand their audiences, heading to a beta in December 2026. You lead the technology in a team of eight.

- **Data.** More than 3 million Facebook and Instagram comments on 111,000+ posts, collected through Meta's official API, with images and videos read through OCR, CLIP and Whisper.
- **Labelling, which is your research question inside a product.** Claude labels about 6,000 posts and writes a reason for each label. Rules mined from those reasons become the prompt for DeepSeek, a cheaper model, which labels about 12,000 more. Those labels train your own classifiers (XLM-R, DistilBERT, DeBERTa). The team hand-labelled about 500 posts and comments to check both models, and you are building evaluation around a human-labelled golden set so no model reaches users unchecked.
- **Hostility, measured carefully.** Toxicity and intent classifiers, a storm detector for sudden bursts of hostile comments, and filters so that dialect and friendly banter are not mistaken for abuse. That last one is a measurement problem in miniature: the same words mean different things in different communities.

- **Scale of the build.** 48 models in the agreed architecture, from comment intent to audience communities, and Art-E, an advisor that answers only from each client's own data, with citations.

Say it in one sentence: "At Fluency and Artemis AI, brands and governments act on machine-labelled numbers that are rarely checked against people; my PhD question is how to check them."

4. What you told CSH (keep answers consistent)



Answer 1: a research project, your role, and what did not go as expected. Your MSc dissertation at Portsmouth (2025, supervisor Dr Ella Haig) asked how public reaction to UK economic policy differed across platforms. It was yours from question to write-up: you obtained ethics approval, collected 255,446 public comments from X, Reddit, YouTube and Quora through official APIs and licensed feeds, and built the whole analysis in Python: VADER and BiLSTM baselines, a fine-tuned BERT classifier, calibration, McNemar and bootstrap tests, chi-square tests across topics, Prophet forecasts, NRC emotions, and LIME and integrated gradients to see what the model relied on. You also hand-labelled 200 posts.

What went wrong, told honestly: BERT scored a macro-F1 of 0.878, but its training labels had been seeded by the VADER lexicon, the score was largely in-sample, and your 200-post check reviewed VADER's own labels, so the agreement (kappa about 0.78) measured agreement with VADER, not truth. The validation was circular: it showed BERT had learned to copy VADER. Underneath was a second untested assumption, that "negative" means the same on Reddit as on YouTube. *What you learned:* an agreement score describes the reference as much as the model, so the first design question is where the yardstick comes from; and a comparison between groups needs evidence that the instrument behaves alike in each, a routine test in survey research that text analysis rarely runs. That became the paper with Dr Haig, and for its application you fixed fifteen policy events in advance, before computing any counts.

Answer 2: a discovery from a CSH group. You chose Chen, Kmetty, Iñiguez and Omodei (2025), "The public that engages invisibly", *Communication Methods and Measures* 19(4): 294–312. By linking a survey of Hungarian internet users to the YouTube histories they donated, they show that people who comment on political videos are more polarised and more selective than people who only watch, so studies built on visible traces such as comments overstate how divided audiences are. *Your question:* is the gap between commenters and watchers stable enough, across platforms, topics and time, that a small donated panel could correct polarisation estimates built from public comments at scale? *Why it matters to you:* it has the same structure as your problem. Public comments are a large, cheap, selected instrument; donated data is a small, expensive, closer criterion. If the relation holds steady, a few hundred donors could calibrate millions of comments, just as a small hand-labelled sample anchors a machine-labelled corpus; if it breaks, knowing where is itself a finding. You meet the same gap at work: brands judge their pages by comments, though most of their audience never comments.

Answer 3: a problem you want to explore. *Statement:* when is a machine-labelled construct (sentiment, stance or a narrative frame) comparable across platforms, annotator models and time, and how should the differences be corrected before we track how narratives spread between platforms? *Why it matters:* research on narratives and polarisation increasingly pools platforms and uses LLMs as annotators, and any difference mixes real change with how each platform elicits expression and how each model reads it, while the models change version underneath us. You cited Gaynor, Miler, Goel, Hoyle and Resnik (2026, *Journal of Politics*):

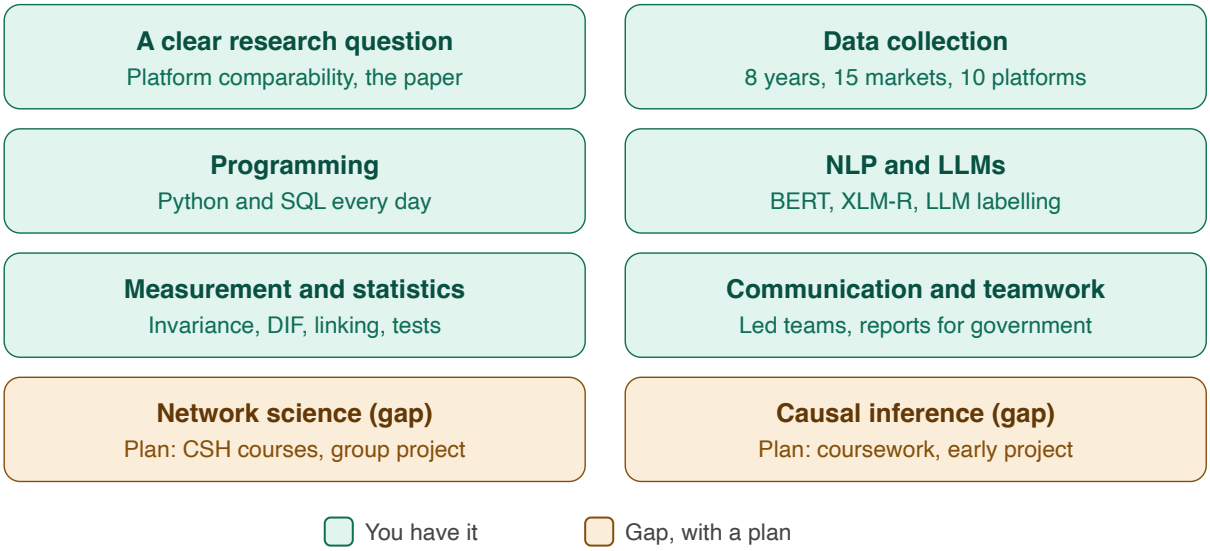
legislators' positions differ across votes, speeches and tweets; Fan et al. (2025) on removing the medium from embeddings by concept erasure; Licht et al. (2025) on LLMs measuring scalar constructs; and Hoyle et al. (2021) on automated metrics drifting from human judgement. *Why tractable, in four steps:* (1) your simulation has known ground truth and runs in seconds, so you can add models and time as facets with no new data; (2) you can build an event-anchored corpus from official APIs, labelled by several LLMs, with a larger hand-labelled sample per platform and a second coder, then split the variance and test each frame for DIF; (3) concept erasure tests whether removing the platform signal also removes real differences; (4) corrected and uncorrected measures go into a network diffusion analysis to show when conclusions change. *For whom:* researchers who pool platforms or switch models mid-study; public bodies, NGOs and journalists reading social-listening dashboards; anyone asking whether a narrative really moved between platforms. *What you would learn at CSH:* the network science of diffusion.

Your portfolio (sent with the CV) listed six pieces, each with why it was there: the paper; the dissertation; the Artemis labelling chain; the Fluency twelve-month study; the self-healing and builder agents ("a collector that breaks silently does not stop a time series, it thins it"); and the multilingual Dubizzle dashboards that predicted revenue and showed what customers said.

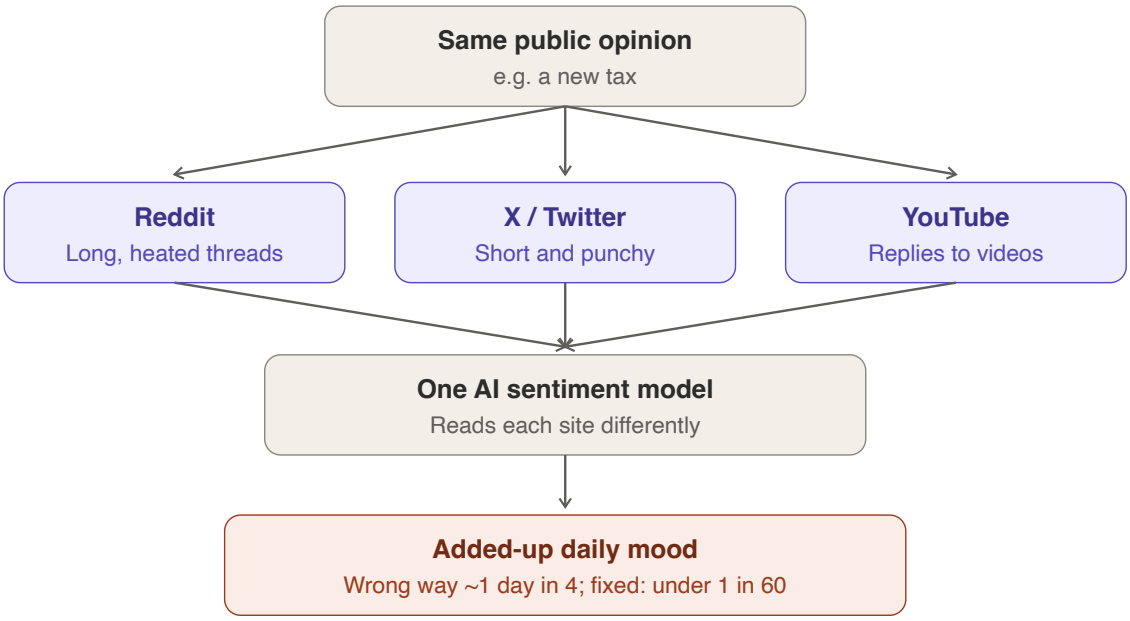
4b. The people and projects you named (verified 19 Oct)

- **Alexander Hoyle**: CSH Digital Innovation School faculty, listed for NLP in social science, measurement and validation, and reasoning in LLMs; PhD Maryland 2024 with Philip Resnik, currently also associated with the ETH AI Center. His thesis is about developing and measuring latent constructs from text with human-centred validation. He is the closest match to your question; say "his work on validating NLP measures".
- **Elisa Omodei**: DIS faculty (computational social science, online social systems, biases and inequalities) and a co-author of the paper you chose for Answer 2. **Gerardo Iñiguez** is her co-author on it.
- **Fariba Karimi**: DIS faculty working on homophily and biases in networks and algorithms, relevant to polarisation and minority visibility.
- **Springboard projects** you referenced, both on CSH's 2027 list: *Understanding narratives at scale* (how narratives spread and shape discourse, with NLP, network science and large text data) and *Polarization and minority representation on online platforms* (how migration and minority-related discourse evolves online and relates to polarisation). Related ones worth knowing: *Designing link recommendation to reduce polarization* and *Identifying causal mechanisms for online behavior*.

5. What a PhD needs, and what you bring



6. Your paper in 20 seconds



“AI reads opinions differently on Reddit than on YouTube, because people write differently on each. So adding platforms together can flip the answer: in my simulation, the overall mood points the wrong way on nearly a quarter of days, and under one day in sixty once each platform is corrected. The paper gives a checklist to test whether a comparison is fair, and a way to fix it.”

7. Practice answers (45–75 seconds each)

Tell us about yourself.
I'm a data engineer and researcher from Pakistan, and I work on social media data. I started web scraping on Fiverr as a computer science student, did more than 300 projects, then built Dubizzle Labs' first scraping system across 15 markets in four languages and was promoted twice. A Chevening Scholarship took me to Portsmouth for an MSc in Data Analytics, where my dissertation compared public reaction to UK economic policy across four platforms. Today I work on social media analysis at M+C Saatchi Fluency in London, for brands and UK government, and I'm co-founder and CTO of Artemis AI. Last week my first paper, with my supervisor, went live on arXiv.

Describe a research project and what went wrong.
Use section 4, Q1: the project, the circular validation you found yourself, the paper it became, the quarter-of-days result, and the lesson about independent yardsticks.

What have you built in industry, and what did it teach you?
Use section 3: one Fluency example (prompt-versioned LLM tagging), one Artemis AI example (explained labels, dialect and banter filters), then the one-sentence thread.

Why a PhD, and why now?
Industry taught me to collect and label online data at scale; it didn't teach me to show rigorously what those numbers say about people. Publishing the paper showed me this is the work I enjoy most, and I want the training to do it properly, then carry it back into how these tools are built.

Why CSH, and why complexity science?
Opinion online is a system, not a pile of posts: people, platforms and algorithms shape each other, and what we can see is a selected slice. Chen and colleagues showed commenters are more polarised than watchers, which is my problem in another form. Alexander Hoyle's validation work is closest to mine, and I want to learn network science and diffusion, which I don't have yet.

What would you work on?
Use section 4, Q3, in three steps: extend the simulation, build an event-anchored corpus labelled by several LLMs with human checks, then show when diffusion conclusions change after correction.

Strengths and gaps?
Use section 5: name two strengths with evidence, then the two gaps and how you will close them.

Teamwork, a setback, five years?
Team: led three at Dubizzle and three at VendueTech, lead the technology in a team of eight at Artemis AI. Setback: the circular validation. Five years: researching how online opinion is measured, so the numbers people act on are tested against people.

Questions for us? (one only)
How are students matched with supervisors in the first year, and is co-supervision across groups common?

8. On the day

- Join at **8:35 pm Lahore**; test Zoom, camera and microphone; quiet room, light in front of you, look at the camera.
- Keep this pack open but do not read from it. Answer the question, then stop. A short pause before answering is fine.
- Same facts as your application: 255,446 comments; team of eight at Artemis AI; arXiv 2610.16757; MSc Distinction, Chevening 2024/25.
- Recording is normal; only object at the start if you do not consent.

9. If you pass: Stage 2 (7–14 January)

You will present a paper chosen by the committee. Past papers: 2026 Clauset, Moore and Newman, “Hierarchical structure and the prediction of missing links in networks” (*Nature* 2008); Couzin et al., “Uninformed individuals promote democratic consensus in animal groups” (*Science* 2011); 2025 Albert, Jeong and Barabási, “Error and attack tolerance of complex networks” (*Nature* 2000); 2024 Watts and Strogatz, “Collective dynamics of small-world networks” (*Nature* 1998). CSH’s advice: read carefully, google, work through the equations or code, and be ready to explain methods, results, conclusions and critiques. Start reading these after Tuesday.

Sources: CSH Graduate Program FAQ (csh.ac.at/education/dis-faq); CSH DIS faculty list (csh.ac.at/education/dis-faculty) and springboard projects (csh.ac.at/education/dis-projects); Crossref record for doi 10.1080/19312458.2025.2542725; CSH invitation email of 14 Oct 2026; AcademicPositions, “The 5 most important questions you’ll be asked during a PhD interview”; Researcher.Life, “PhD admission interview questions and how to answer them”; your CSH answers and CV (13 Oct 2026).